

Chapter 6 Warm-Up: Functions and a Sawtooth Wave

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1 Defining a Function

1. Python functions allow us to give a name to a calculation and evaluate it for different arguments. Begin with

```
import numpy as np
import matplotlib.pyplot as plt
```

```
def wave(x):
    return np.sin(x)
```

Create 400 values between $-\pi$ and π :

```
x = np.linspace(-np.pi,
                np.pi, 400)
```

```
y = wave(x)
```

Here, the argument x is an array, so the function evaluates $\sin(x)$ at every value in the array.

2 Adding Trigonometric Terms

1. The periodic sawtooth function

$$f(x) = x, \quad -\pi < x < \pi$$

can be approximated by the sum

$$f_N(x) = 2 \sum_{n=1}^N \frac{(-1)^{n+1}}{n} \sin(nx).$$

Define a Python function that calculates this sum:

```
def sawtooth(x, N):
    total = 0.0

    for n in range(1, N + 1):
        term = 2 * (-1)**(n + 1)
        term = term * np.sin(n*x)/n
        total = total + term

    return total
```

- What are the two parameters of this function?
- What does `n` represent?
- Why must `total` be defined before the loop?
- What value is returned by the function?

3 Code Lab: Build the Sawtooth

1. Evaluate the approximation with only one term:

```
y1 = sawtooth(x, 1)
```

Now evaluate it using more terms:

```
y3 = sawtooth(x, 3)
y10 = sawtooth(x, 10)
```

Plot all three results:

```
plt.plot(x, y1, label="N = 1")
plt.plot(x, y3, label="N = 3")
plt.plot(x, y10, label="N = 10")
```

```
plt.xlabel("x")
plt.ylabel("f(x)")
plt.legend()
plt.grid()
plt.show()
```

- Which approximation looks most like a straight line between $-\pi$ and π ?
- What happens near $x = -\pi$ and $x = \pi$?
- Increase the number of terms to

`N = 50`

and plot the result. Does adding terms make the transition at the edge sharper?

4 Challenge

1. Modify the plotting code so that it compares

$$N = 1, \quad N = 5, \quad N = 20.$$

Then add the exact line $f(x) = x$:

```
plt.plot(x, x, label="f(x) = x")
```

Which value of N gives the closest approximation over most of the interval?